



DESIGN AND ANALYSIS OF ALGORITHMS

Branch and Bound

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- Dynamic Programming
 - ▶ Computing a Binomial Coefficient
 - ▶ The Knapsack Problem
 - ▶ Memory Functions
 - ▶ Warshall's and Floyd's Algorithms
 - ▶ Optimal Binary Search Trees
- Limitations of Algorithmic Power
 - ▶ Lower-Bound Arguments
 - ▶ Decision Trees
 - ▶ P, NP, and NP-Complete, NP-Hard Problems
- Coping with the Limitations
 - ▶ Backtracking
 - ▶ **Branch and Bound**

Concepts covered

- Backtracking
 - ▶ General Approach
 - ▶ Knapsack Problem
 - ▶ Assignment Problem
 - ▶ Travelling Salesman Problem

- An enhancement of backtracking
- Applicable to optimization problems
- For each node (partial solution) of a state-space tree, computes a bound on the value of the objective function for all descendants of the node (extensions of the partial solution)
- Uses the bound for:
 - ▶ ruling out certain nodes as “nonpromising” to prune the tree (if a node’s bound is not better than the best solution seen so far)
 - ▶ guiding the search through state-space

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Example: Assignment Problem

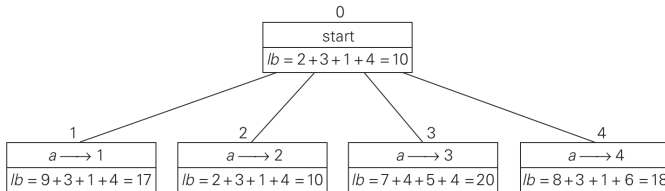
- Select one element in each row of the cost matrix C so that:
 - ▶ no two selected elements are in the same column
 - ▶ the sum is minimized
- Example

	Job 1	Job 2	Job 3	Job 4
Person a	9	2	7	8
Person b	6	4	3	7
Person c	5	8	1	8
Person d	7	6	9	4

- Lower bound (sum of smallest elements in each row): $2 + 3 + 1 + 4 = 10$
- Best-first branch-and-bound variation: Generate all the children of the most promising node

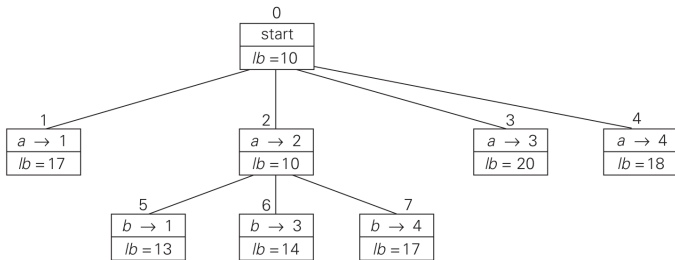
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Example: First two levels of the state-space tree



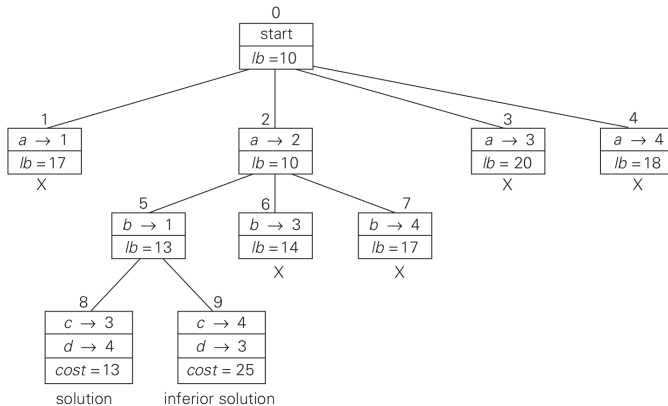
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Example: First three levels of the state-space tree



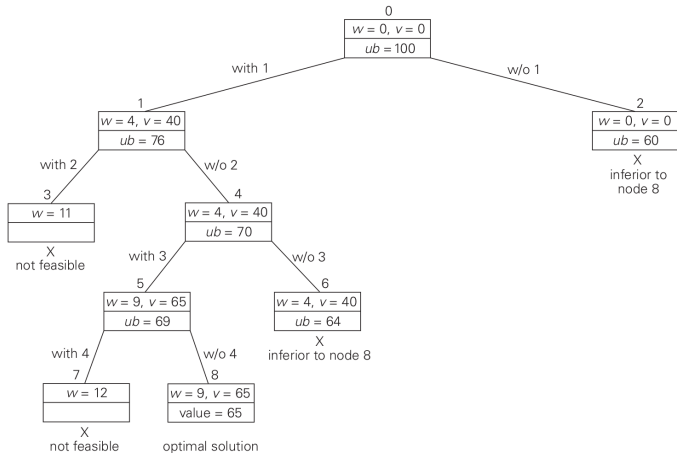
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Example: Complete state-space tree



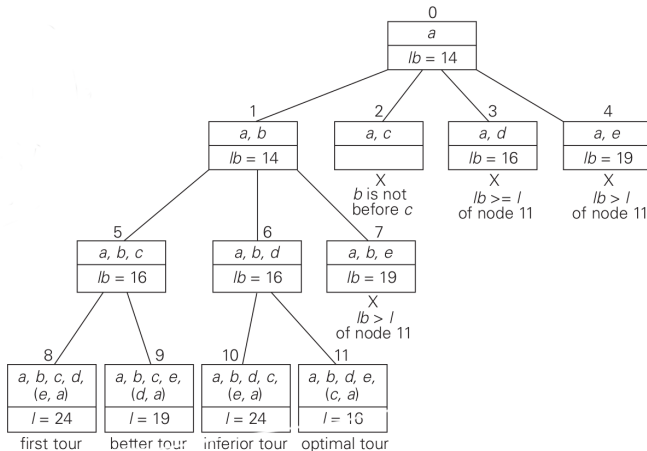
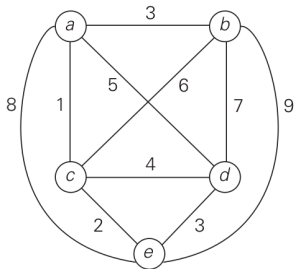
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Example: Knapsack Problem



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Example: Traveling Salesman Problem



- What data structure would you use to keep track of live nodes in a best-first branch-and-bound algorithm?
- Solve the assignment problem by the best-first branch-and-bound algorithm with the bounding function based on matrix columns rather than rows